Electronic and Telecommunication Engineering
University of Moratuwa, Sri Lanka

EN4730 - Convex Engineering Design

Assignment 01

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EN4730 Convex Engineering Design Assignment 01

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1. Determine the following functions and sets are convex, concave or neither convex nor concave with a justification.

   (a) $f(x) = -x\log(x)$.

   (b) $f(\boldsymbol{x}) = e^{x_1} + e^{x_2} + x_1^2$.

   (c) $f(\boldsymbol{x}) = 2x_1^2 - 6x_1x_2 + 2x_2^2$.

   (d) $f(\boldsymbol{x}) = \frac{1}{2}\left(ax_1^2 + bx_2^2\right)$ with $a,\ b \le 0$.

   (e) $f(\boldsymbol{x}) = \frac{1}{2}\left(ax_1^2 + bx_2^2\right)$ with $a \times b > 0$.

   (f) $f(x) = \dfrac{x^\beta - 2}{\beta}$.

   (g) $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \ge 0,\ x_2 \ge 0,\ 2x_1 + x_2 \ge 2,\ x_1 + 2x_2 \le 6\}$.

   (h) $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \ge 0,\ x_2 \ge 0,\ 2x_1^2 + x_2 \ge 2,\ x_1 + 2x_2 \le 4\}$.

   (i) $S = \{(x_1, x_2) \in \mathbb{R}^2 \mid x_1 \ge 0,\ x_2 \ge 0,\ x_1^2 + x_2^2 < 16,\ x_1 + 2x_2 < 4\}$.

   (j) $S = \{(x_1, x_2, a) \in \mathbb{R}^3 \mid x_1 \ge 0,\ x_2 \ge 0,\ x_1^2 + x_2^2 = a^2\}$.

[15 marks]

a) $f(x) = -x\log(x)$

$\text{dom}(f) = x > 0 \Rightarrow$ convex set

$f'(x) = -x \cdot \frac{1}{x} - \log(x) \cdot 1$

$f'(x) = -1 - \log(x)$

$f''(x) = \frac{-1}{x}$

$x > 0 \Rightarrow f''(x) < 0$

$\therefore f(x)$ is concave on $x > 0$.

b) $f(x_1, x_2) = e^{x_1} + e^{x_2} + x_1^2$

$\text{dom}(f) = (x_1, x_2) \in \mathbb{R} \Rightarrow$ convex set

$$\frac{df}{dx_1} = e^{x_1} + 2x_1 \qquad \frac{\partial f}{\partial x_2} = e^{x_2}$$

$$\frac{\partial^2 f}{\partial x_1^2} = e^{x_1} + 2 \qquad \frac{\partial^2 f}{\partial x_2^2} = e^{x_2}$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0 \qquad \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0$$

$$H = \begin{bmatrix} e^{x_1}+2 & 0 \\ 0 & e^{x_2} \end{bmatrix}$$ Hessian Matrix

since H is a diagonal matrix

$\lambda_1 = e^{x_1} + 2$

$\lambda_2 = e^{x_2}$

$\lambda_1 > 0 \qquad \lambda_2 > 0$

since all eigen values > 0,

Hessian matrix is Positive Definite

$\therefore f(x_1, x_2)$ is convex on $\mathbb{R}^2$

c) $f(x_1, x_2) = 2x_1^2 - 6x_1 x_2 + 2x_2^2$

$\text{dom}(f) = (x_1, x_2) \in \mathbb{R} \Rightarrow$ convex set

$$\frac{\partial f}{\partial x_1} = 4x_1 - 6x_2 \qquad \frac{\partial f}{\partial x_2} = -6x_1 + 4x_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = 4 \qquad \frac{\partial^2 f}{\partial x_2^2} = 4$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = -6 \qquad \frac{\partial^2 f}{\partial x_2 \partial x_1} = -6$$

$$H = \begin{bmatrix} 4 & -6 \\ -6 & 4 \end{bmatrix}$$

$$|H - \lambda I| = 0$$

$$\begin{vmatrix} 4-\lambda & -6 \\ -6 & 4-\lambda \end{vmatrix} = 0$$

$$(4-\lambda)^2 - (-6)^2 = 0$$

$$\lambda^2 - 8\lambda - 20 = 0$$

$$\lambda = 10 \qquad \lambda = -2$$

Since H has positive and negative eigen values, H is indefinite.

$f(x_1, x_2)$ is neither convex nor concave.

d) $f(x_1, x_2) = \frac{1}{2}(ax_1^2 + bx_2^2)$ $\quad a, b \leq 0$

$\text{dom}(f) = (x_1, x_2) \in \mathbb{R} \Rightarrow$ convex set

$$\frac{\partial f}{\partial x_1} = ax_1 \qquad \frac{\partial f}{\partial x_2} = bx_2$$

$$\frac{\partial^2 f}{\partial x_1^2} = a \qquad \frac{\partial^2 f}{\partial x_2^2} = b$$

$$\frac{\partial^2 f}{\partial x_1 \partial x_2} = 0 \qquad \frac{\partial^2 f}{\partial x_2 \partial x_1} = 0$$

$$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

since H is a diagonal matrix,

$\lambda_1 = a \qquad \lambda_2 = b$

$\lambda_1 \leq 0 \qquad \lambda_2 \leq 0$

both eigen values ≤ 0

Hessia matrix is negative semi definite

$f(x_1, x_2)$ is concave on $\mathbb{R}^2$.

e) $f(x_1, x_2) = \frac{1}{2}(ax_1^2 + bx_2^2)$ $\quad ab > 0$

$dom(f) = (x_1, x_2) \in \mathbb{R} \Rightarrow$ convex set

$$H = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}$$

since H is a diagonal matrix,

$\lambda_1 = a$ $\quad \lambda_2 = b$

since $a \times b > 0$

Both eigen values have same sign.

<u>case 1</u> $\quad \lambda_1 > 0 \quad \lambda_2 > 0$

Hessian matrix is positive definite.

$f(x_1, x_2)$ is convex on $\mathbb{R}^2$

<u>case 2</u> $\quad \lambda_1 < 0 \quad \lambda_2 < 0$

Hessian matrix is negative definite

$f(x_1, x_2)$ is concave on $\mathbb{R}^2$

f) $f(x) = \dfrac{x^{\beta} - 2}{\beta}$

$f'(x) = x^{\beta-1}$

$f''(x) = (\beta-1)\, x^{\beta-2}$

<u>case1</u> $x > 0$

<u>subcase1</u> $\beta > 1$

$f''(x) > 0 \quad \forall\, x > 0$

$\therefore$ f is convex.

<u>subcase 2</u> $\beta = 1$

$f''(x) = 0$

$f(x) = x - 2$ linear equation

$\therefore$ f is both convex and concave.

<u>subcase 3</u> $\beta < 1$ $(\beta \neq 0)$

$f''(x) < 0 \quad \forall\, x > 0$

$\therefore$ f is concave.

case 2 $\quad x < 0$

subcase 1 $\quad$ even β, $\beta > 1$

$f''(x) > 0 \Rightarrow f$ is convex

subcase 2 $\quad$ even β, $\beta < 1$ $(\beta \neq 0)$

$f''(x) < 0 \Rightarrow f$ is concave

subcase 3 $\quad$ odd β, $\beta > 1$

$f''(x) < 0 \Rightarrow f$ is concave

subcase 4 $\quad$ odd β, $\beta < 1$ $(\beta \neq 0)$

$f''(x) > 0 \Rightarrow f$ is convex

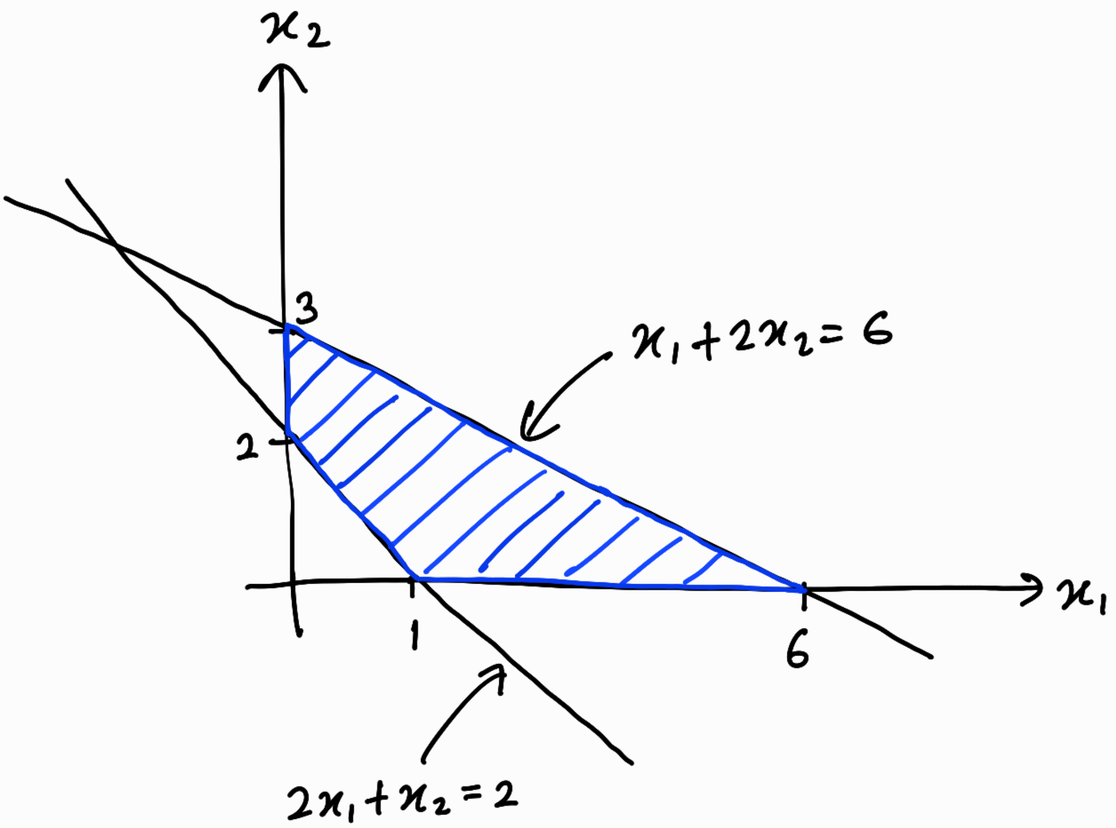
subcase 5 $\quad \beta = 1$

$f(x) = x - 2$ $\quad$ linear equation

$\therefore f$ is both convex and concave

$f(x)$ is not globally convex or concave on $\mathbb{R}$.

g) $S = \{ (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0, 2x_1 + x_2 \geq 2, x_1 + 2x_2 \leq 6 \}$



all constraints are linear inequalities.

they form closed half spaces. (C_i)

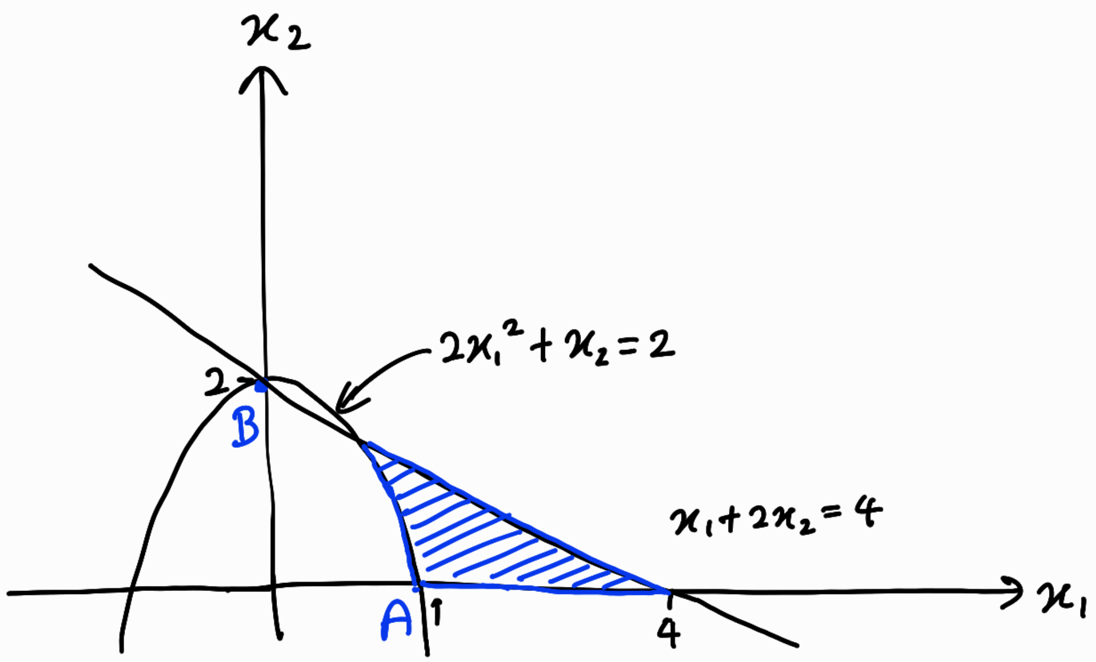
they are convex sets.

$$S = C_1 \cap C_2 \cap C_3 \cap C_4$$

Intersection of convex sets is also convex.

$\therefore$ S is convex set.

h) $S = \{ (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0, x_2 \geq 0,$
$2x_1^2 + x_2 \geq 2, \; x_1 + 2x_2 \leq 4 \}$



Let's take $A = (1, 0)$

$1 \geq 0 \qquad 0 \geq 0 \qquad 2 + 0 \geq 2 \qquad 1 + 0 \leq 4$

It satisfies all constraints

It's in set S.

Now take $B = (0, 2)$

$0 \geq 0 \qquad 2 \geq 0 \qquad 2(0) + 2 \geq 2 \qquad 0 + 2(2) \leq 4$

It satisfies all constraints

It's in set S.

Midpoint between A and B,

$M = (0.5, 1)$

Let's check whether it satisfy $2x_1^2 + x_2 \geqslant 2$

$2(0.5)^2 + 1 \geqslant 2$

$1.5 \geqslant 2$

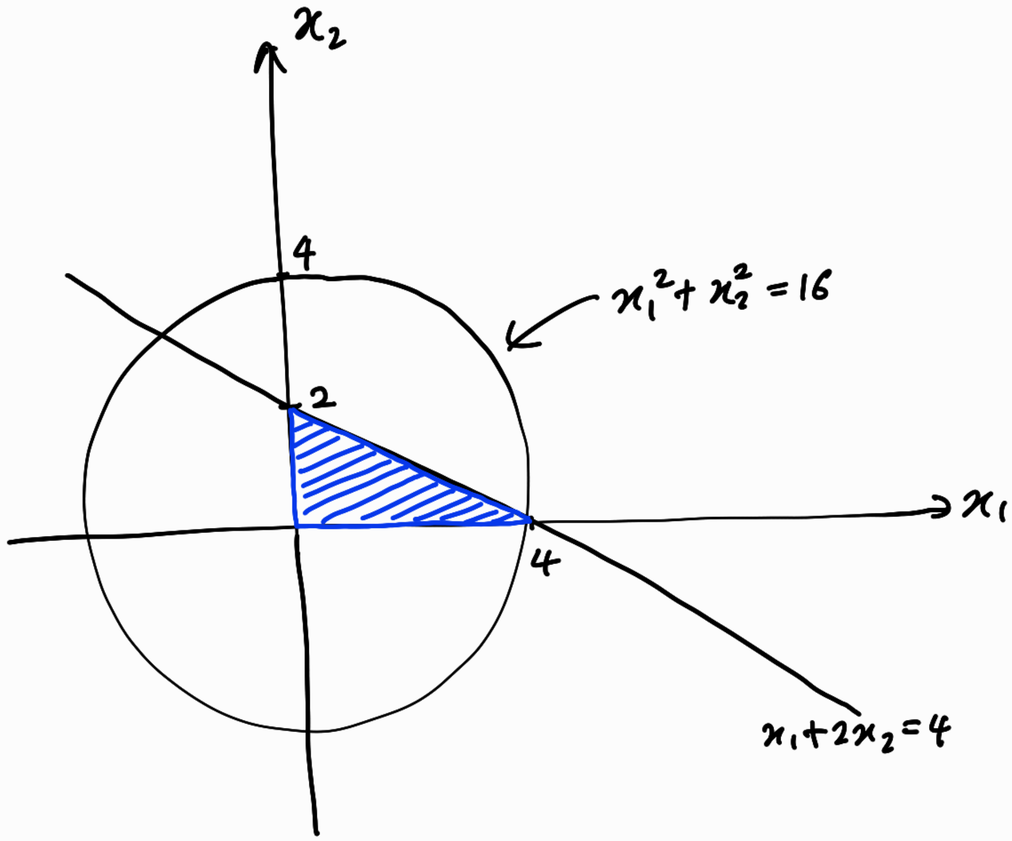
It's false.

Midpoint is not in set S.

$\therefore$ S is not convex set.

i) $S = \{ (x_1, x_2) \in \mathbb{R}^2 \mid x_1 \geq 0,\ x_2 \geq 0,$

$x_1^2 + x_2^2 < 16,\ x_1 + 2x_2 \leq 4 \}$



$x_1 \geq 0$, $x_2 \geq 0$, $x_1 + 2x_2 \leq 4$ are linear inequalities.

they form closed half spaces. (C_i)

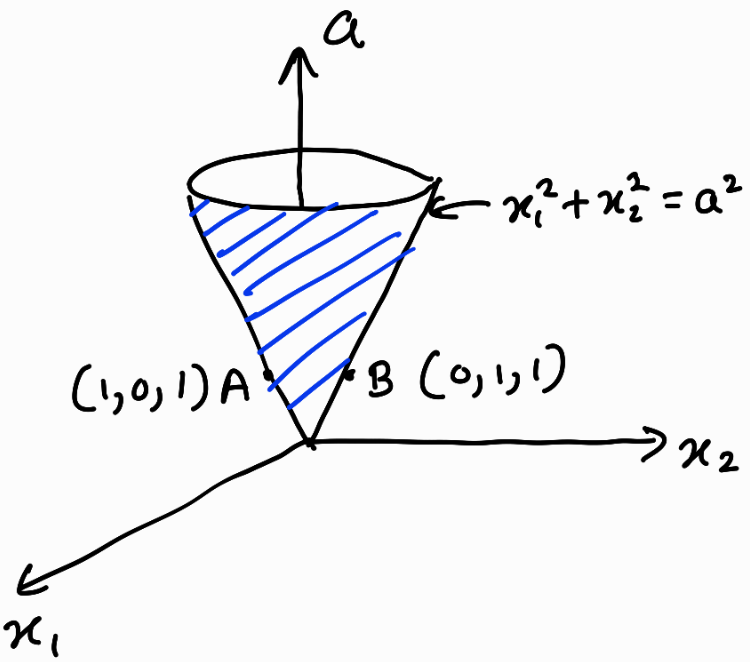
they are convex sets.

$S = C_1 \cap C_2 \cap C_3$

Intersection of convex sets is also convex.

$\therefore S$ is convex set.

j) $S = \{ (x_1, x_2, a) \in \mathbb{R}^3 \mid x_1 \geq 0 , x_2 \geq 0, x_1^2 + x_2^2 = a^2 \}$



Let's take point $A = (1,0,1)$

$1 \geq 0 \quad 0 \geq 0 \quad 1 + 0 = 1$

since it satisfies all constraints

$A = (1,0,1)$ is in set S.

Let's take point $B = (0,1,1)$

$0 \geq 0 \quad 1 \geq 0 \quad 0 + 1 = 1$

since it satisfies all constraints

$B = (0,1,1)$ is in set S.

Midpoint between A and B,

$$M = (0.5,\ 0.5,\ 1)$$

Let's check whether M satisfies

$$x_1^2 + x_2^2 = a^2$$

$$(0.5)^2 + (0.5)^2 = 1^2$$

$$0.5 = 1$$

It's false.

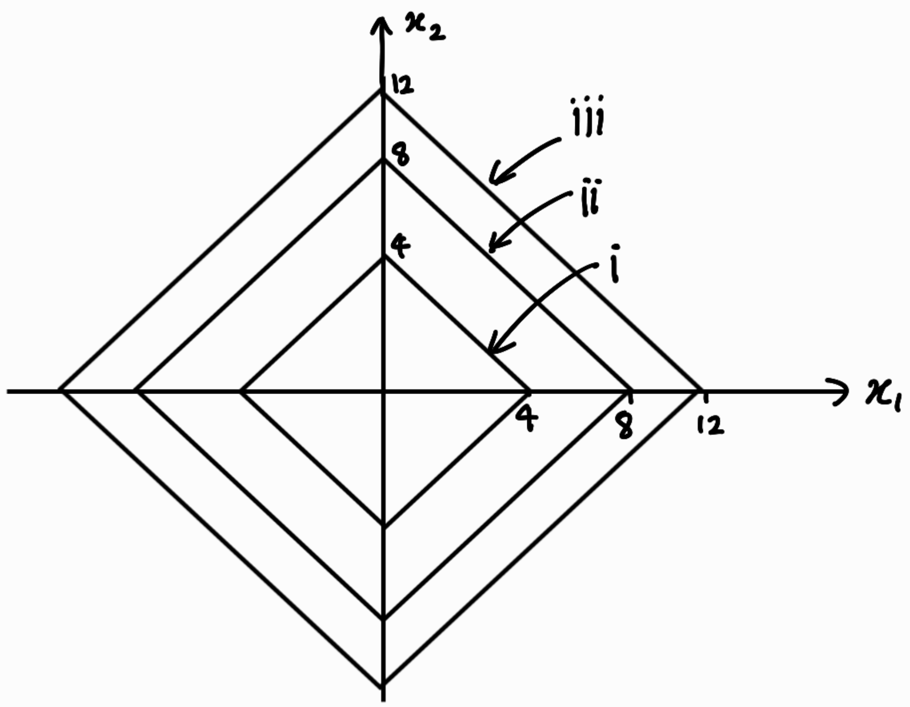
Midpoint is not in set S.

$\therefore$ S is not convex set.

② a) i) $\|x\|_1 = 4$ $\quad |x_1| + |x_2| = 4$

ii) $\|x\|_1 = 8$ $\quad |x_1| + |x_2| = 8$

iii) $\|x\|_1 = 12$ $\quad |x_1| + |x_2| = 12$



b)

$$x^T a = c$$

$$\begin{bmatrix} x_1 & x_2 \end{bmatrix} \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} = c$$

$$a_1 x_1 + a_2 x_2 = c$$

This is a straight line.

Unique solution occurs when line crosses $\|x\|_1$ at a vertex.

line can't be parallel to any edge of $\|x\|_1$

$$\|x\|_1 \text{ slope} = +1/-1$$

$$\text{line slope} = -\frac{a_1}{a_2}$$

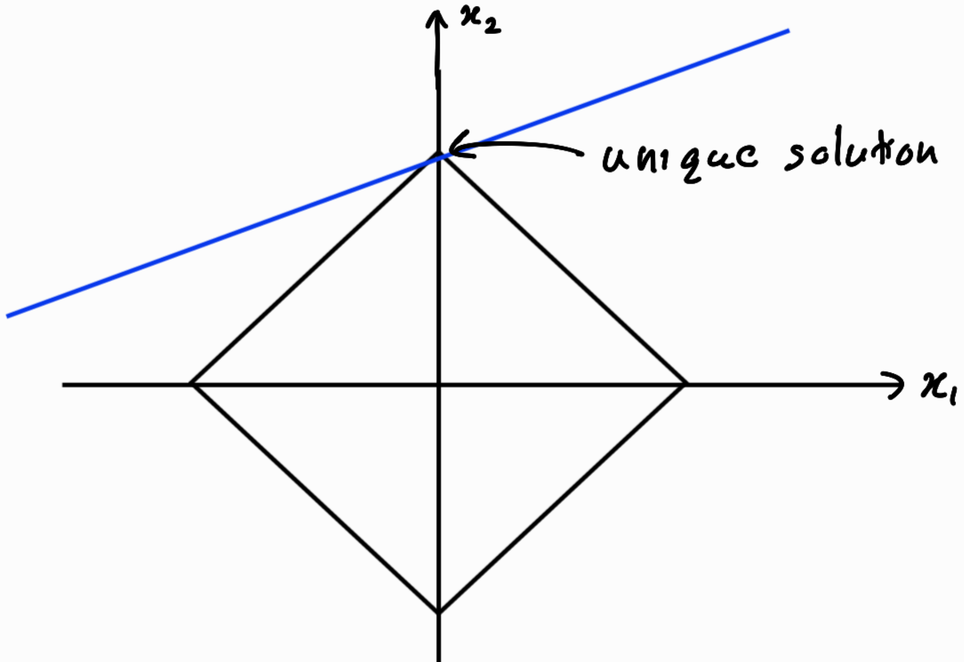
for unique solution,

$$\|x\|_1 \text{ slope} \neq \text{Constraint line slope}$$

$$-\frac{a_1}{a_2} \neq \pm 1$$

$$a_1 \neq \pm a_2$$

$$|a_1| \neq |a_2|$$



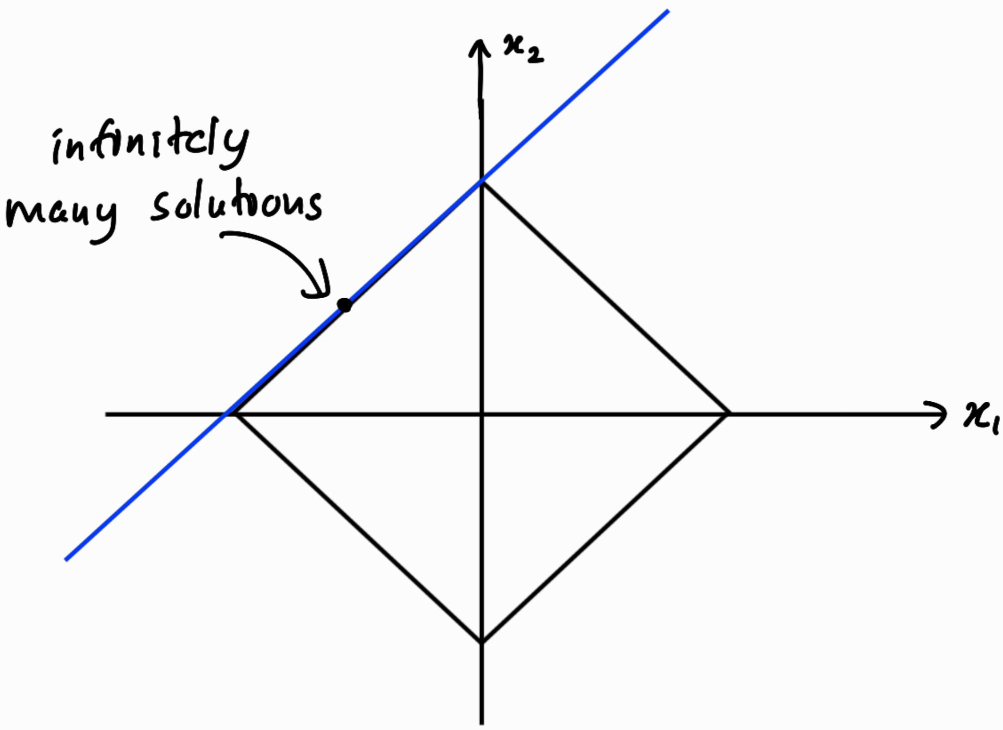
c) for infinitely many solutions

$\|x\|_1$ slope = Constraint line slope

$$\frac{-a_1}{a_2} = \pm 1$$

$$a_1 = \pm a_2$$

$$|a_1| = |a_2|$$



d) unique solution $|a_1| \neq |a_2|$

$$(a_1, a_2) = (1, 2)$$

infinitely many solutions $|a_1| = |a_2|$

$$(a_1, a_2) = (1, 1)$$

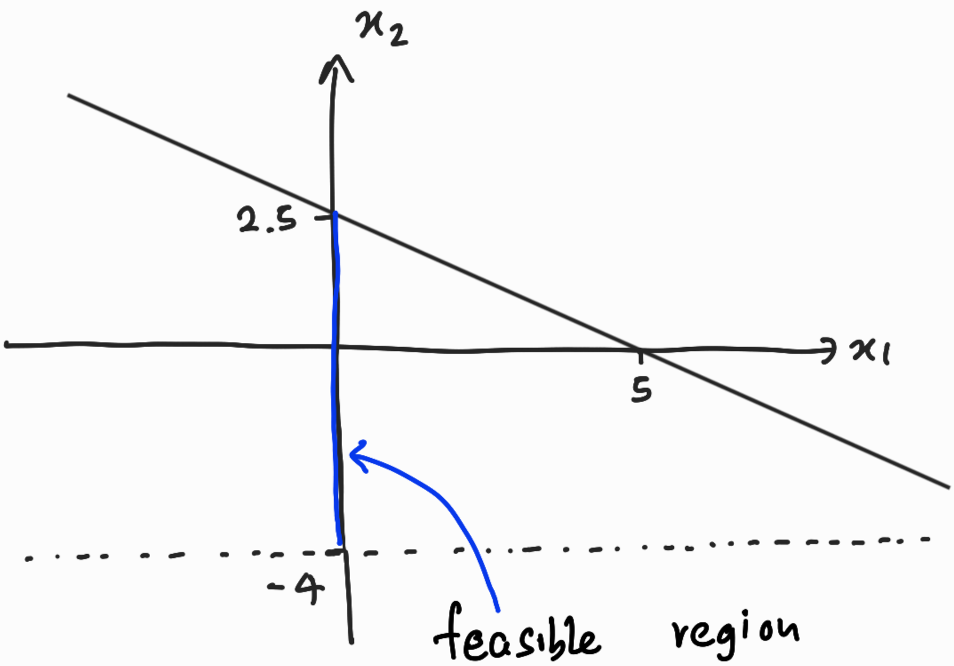
③ a) $Ax \leq c$

$$\begin{bmatrix} -1 & 0 \\ 1 & 0 \\ 0 & -1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \\ 4 \\ 5 \end{bmatrix}$$

$$\begin{bmatrix} -x_1 \\ x_1 \\ -x_2 \\ x_1 + 2x_2 \end{bmatrix} \leq \begin{bmatrix} 0 \\ 0 \\ 4 \\ 5 \end{bmatrix}$$

$$\left.\begin{matrix} -x_1 \leq 0 & \quad x_1 \leq 0 \\ x_1 \geq 0 & \end{matrix}\right\} \quad x_1 = 0$$

$$\left.\begin{matrix} -x_2 \leq 4 & \quad x_1 + 2x_2 \leq 5 \\ x_2 \geq -4 & \quad x_2 \leq 2.5 \end{matrix}\right\} \quad -4 \leq x_2 \leq 2.5$$



$(x_1 = 0$ and $-4 \leqslant x_2 \leqslant 2.5)$

line segment on x_2 axis between $(0,-4)$ and $(0,2.5)$

b) i) $f(x) = x_2$

$f(x)$ has only one minimum at $(0,-4)$

$f(x)$ min $= -4$

$f(x)$ has a unique solution in feasible region.

ii) $f(x) = x_2^2$

$f(x)$ has only one minimum at $(0,0)$

$f(x)$ min $= 0$

$f(x)$ has a unique solution when either $x_1 = 0$ or $x_2 = 0$.